Supplementary material to
An Informational Theory of Privacy

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1. Extension: State matching

In this section, we consider a model where the private information of citizens in the information aggregation stage is not directly their personal payoff of policy $p = 1$. Instead citizens have all the same payoff of policy $p = 1$ but each citizen only receives a noisy signal of this payoff. This has a striking implication: Chilling makes every citizen worse off. The reason is that chilling inhibits information aggregation. In the main paper citizens have private preferences over outcomes and therefore some citizens (those with negative θ_i) gain from chilling. Since all citizens have the same interest – implementing the policy if and only if the common payoff consquence is positive, – everyone looses in this setup from chilling.

More precisely, the setting is as follows: The state of the world θ is distributed standard normally and this θ is the payoff consequence of policy $p = 1$ for each citizen. However, the realization of θ is unknown. Each citizen obtains a private signal θ_i which is normally distributed around the true state θ, i.e. $\theta_i \sim N(\theta, \sigma^2)$ where we denote the cdf by $\tilde{\Phi}_\theta$ and the pdf by $\tilde{\phi}_\theta$. All θ_i are assumed to be independent draws from this distribution. The interaction type of citizen i, τ_i, is drawn from Γ_{θ_i} where again $\Gamma_{\theta_i'}$ is assumed to first order stochastically dominate $\Gamma_{\theta_i''}$ if and only if $\theta_i' > \theta_i''$. This creates a positive correlation between θ_i and τ_i. The interaction stage is exactly the same as in the model of the main paper. That is, without privacy a strategy for OP states which of the two actions A and M OP plays against a citizen who chose $p_i = 0$ and which against a citizen who chose $p_i = 1$. With privacy, OP only decides which of the two actions he chooses against all citizens. This means that – to keep the setting comparable to the main paper – we do not consider strategies (or beliefs) that are contingent upon the number of citizens choosing $p_i = 1$. This is a simplification. However, one can easily imagine settings where OP has to commit to a strategy before he gets to know the citizens' p_is. This is, for example, the case if the interaction is between i and an agent representing OP and p_i is only learned in the interaction. OP then has to instruct the agent in advance how to act.

The main change is, therefore, that citizen i's payoff is $\theta p - \mathbb{1}_{s(p_i)=A}\delta(\tau_i)$; that is, θ instead of θ_i enters the utility function. To keep the model tractable, we will assume that $q(m/n) = m/n$.

We first replicate some intermediary results from the main text in this modified setting.

Lemma 1. *For citizens, only cutoff strategies $t(\tau_i)$ are rationalizable. In the privacy case, the optimal cutoff is $t^p(\tau_i) = 0$ for all τ_i.*

Proof. If citizen i receives signal θ_i, he updates his belief α about θ according to Bayes' rule yielding

$$\alpha(\theta'|\theta_i) = prob(\theta \leq \theta'|\theta_i) = \frac{\int_{-\infty}^{\theta'} \tilde{\phi}_\theta(\theta_i)\, d\Phi(\theta)}{\int_{\mathbb{R}} \tilde{\phi}_\theta(\theta_i)\, d\Phi(\theta)}.$$

From the normality assumptions, it follows that the pdf of the belief is single peaked with its peak between 0 (the mean of the prior) and θ_i. Furthermore, $\mathbb{E}[\theta|\theta_i] = \int_{\mathbb{R}} \theta\, d\alpha(\theta|\theta_i)$ is strictly increasing in θ_i with

limits $\lim_{\theta_i \to \infty} = \infty$ and $\lim_{\theta_i \to -\infty} = -\infty$. To see this, note that

$$\begin{aligned}
\mathbb{E}[\theta|\theta_i] &= \int_{\mathbb{R}} \theta \frac{\tilde{\phi}_\theta(\theta_i)\phi(\theta)}{\int_{\mathbb{R}} \tilde{\phi}_{\hat{\theta}}(\theta_i)\, d\Phi(\hat{\theta})}\, d\theta \\
&= \frac{\int_{\mathbb{R}} \theta e^{-(\theta_i-\theta)^2/(2\sigma^2)} e^{-\theta^2/2} d\theta}{\int_{\mathbb{R}} e^{-(\theta_i-\theta)^2/(2\sigma^2)} e^{-\theta^2/2} d\theta} \\
&= \frac{\theta e^{-(-2\theta_i\theta+\theta^2(1+\sigma^2))/(2\sigma^2)} d\theta}{\int_{\mathbb{R}} e^{-(-2\theta_i\theta+\theta^2(1+\sigma^2))/(2\sigma^2)} d\theta} \\
&= \frac{\frac{1}{\sqrt{2\pi}\sigma/(\sqrt{1+\sigma^2})} \int_{\mathbb{R}} \theta e^{-\frac{\theta_i^2/(1+\sigma^2)^2 - 2\theta_i\theta/(1+\sigma^2)+\theta^2}{2\sigma^2/(1+\sigma^2)}}\, d\theta}{\frac{1}{\sqrt{2\pi}\sigma/(\sqrt{1+\sigma^2})} \int_{\mathbb{R}} e^{-\frac{\theta_i^2/(1+\sigma^2)^2 - 2\theta_i\theta/(1+\sigma^2)+\theta^2}{2\sigma^2/(1+\sigma^2)}}\, d\theta} \\
&= \frac{\theta_i}{1+\sigma^2}
\end{aligned}$$

where the last equality holds as the numerator of the second but last line is the expected value of a randome variable distributed $N(\theta_i/(1+\sigma^2), \sigma^2/(1+\sigma^2)^2)$ and the denominator of the second but last line is simply 1 (as it integrates over the density of this random variable).

Citizen i's expected payoff difference between choosing $p_i = 1$ and $p_i = 0$ is[1]

$$-\delta(\tau_i)\Delta + \mathbb{E}[\theta|\theta_i]/n = -\delta(\tau_i)\Delta + \frac{\theta_i}{(1+\sigma^2)n} \tag{1}$$

where Δ is again the difference between the probabilities that OP plays A against citizens with $p_i = 1$ and citizens with $p_i = 0$. Clearly, it is optimal to play $p_i = 0$ ($p_i = 1$) for sufficiently low (high) θ_i. (Note that $max_{\tau_i \in [\underline{\tau}, \bar{\tau}]} \delta(\tau_i)$ is bounded.) Furthermore, $\mathbb{E}[\theta|\theta_i]$ is strictly increasing in θ_i which implies that i's best response is a cutoff strategy. Consequently, only cutoff strategies are best responses. The optimal cutoff is given by $t(\tau_i) = (1+\sigma^2)n\delta(\tau_i)\Delta$.

In the privacy case, $\Delta = 0$ and therefore the optimal cutoff is $t^p(\tau_i) = 0$. □

OP's belief over τ_i given p_i is given by

$$\begin{aligned}
\beta_0(\tau') = prob(\tau \le \tau' | p_i = 0) &= \frac{\int_{\mathbb{R}} \int_{\mathbb{R}} \int_{\underline{\tau}}^{\tau'} \mathbb{1}_{t(\tau_i) \ge \theta_i}\, d\Gamma_{\theta_i}(\tau_i)\, d\tilde{\Phi}_\theta(\theta_i)\, d\Phi(\theta)}{\int_{\mathbb{R}} \int_{\mathbb{R}} \int_{\underline{\tau}}^{\bar{\tau}} \mathbb{1}_{t(\tau_i) \ge \theta_i}\, d\Gamma_{\theta_i}(\tau_i)\, d\tilde{\Phi}_\theta(\theta_i)\, d\Phi(\theta)} \\
\beta_1(\tau') = prob(\tau \le \tau' | p_i = 1) &= \frac{\int_{\mathbb{R}} \int_{\mathbb{R}} \int_{\underline{\tau}}^{\tau'} \mathbb{1}_{t(\tau_i) \le \theta_i}\, d\Gamma_{\theta_i}(\tau_i)\, d\tilde{\Phi}_\theta(\theta_i)\, d\Phi(\theta)}{\int_{\mathbb{R}} \int_{\mathbb{R}} \int_{\underline{\tau}}^{\bar{\tau}} \mathbb{1}_{t(\tau_i) \le \theta_i}\, d\Gamma_{\theta_i}(\tau_i)\, d\tilde{\Phi}_\theta(\theta_i)\, d\Phi(\theta)}.
\end{aligned}$$

OP's expected utility of playing A against a citizen choosing policy $p_i = 0$ or $p_i = 1$ are then

$$\begin{aligned}
v_0 &= \int_{\mathbb{R}} \tau\, d\beta_0(\tau) \\
v_1 &= \int_{\mathbb{R}} \tau\, d\beta_1(\tau).
\end{aligned}$$

Lemma 2. *In every perfect Bayesian equilibrium (without privacy), $v_1 \ge v_0$.*

Proof. Suppose otherwise. Then $\Delta < 0$ which implies that $t(\tau_i)$ is decreasing. Denote the inverse of t by z. From OP's point of view θ_i is distributed according to the cdf

$$\hat{\Phi}(\theta_i) = \int_{\mathbb{R}} \tilde{\Phi}_\theta(\theta_i)\, d\Phi(\theta).$$

[1] Recall that $q(m/n) = m/n$ which means that i's "influence" on the policy decision is $1/n$.

Using this distribution $\hat{\Phi}$ we can replicate the proof from the main paper one-to-one:

$$\begin{aligned}
v_1 &= \frac{\int_{t(\bar{\tau})}^{t(\underline{\tau})} \int_{z(\theta_i)}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i) + \int_{t(\underline{\tau})}^{\infty} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i)}{\int_{t(\bar{\tau})}^{t(\underline{\tau})} \int_{z(\theta_i)}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i) + \int_{t(\underline{\tau})}^{\infty} \int_{\underline{\tau}}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i)} \\
&\geq \frac{\int_{t(\bar{\tau})}^{\infty} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i)}{\int_{t(\bar{\tau})}^{\infty} \int_{\underline{\tau}}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i)} \\
&> \frac{\int_{-\infty}^{t(\underline{\tau})} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i)}{\int_{-\infty}^{t(\underline{\tau})} \int_{\underline{\tau}}^{\bar{\tau}} d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i)} \\
&\geq \frac{\int_{-\infty}^{t(\bar{\tau})} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i) + \int_{t(\bar{\tau})}^{t(\underline{\tau})} \int_{\underline{\tau}}^{z(\theta_i)} \tau \, d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i)}{\int_{-\infty}^{t(\bar{\tau})} \int_{\underline{\tau}}^{\bar{\tau}} \tau \, d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i) + \int_{t(\bar{\tau})}^{t(\underline{\tau})} \int_{\underline{\tau}}^{z(\theta_i)} \tau \, d\Gamma_{\theta_i}(\tau) \, d\hat{\Phi}(\theta_i)} \\
&= v_0
\end{aligned}$$

which contradicts our starting point $v_1 < v_0$. □

The previous result implies that $\Delta \geq 0$ and therefore $t^{np}(\tau_i) = (1+\sigma^2)n\delta(\tau_i)\Delta \geq 0 = t^p(\tau_i)$. We therefore get chilling.

Proposition 1. *The equilibrium cutoff of a type τ_i is higher without privacy than with privacy. If the absence of privacy affects OP's behavior, this relation is strict. The difference of equilibrium cutoffs with and without privacy is increasing in τ_i.*

To establish that this chilling indeed hurts every citizen – as we claimed above – we have to show that the privacy cutoff zero leads to a higher expected consumer surplus than $t^{np}(\tau) > 0$.

Lemma 3. *The cutoff strategy $t^p(\tau) = 0$, i.e. the equilibrium strategy of the privacy case, gives a higher expected consumer surplus in the information aggregation stage than any other $t^{np}(\tau) > 0$.*

Proof. Let $t(\tau)$ be the strategy maximizing expected consumer welfare. Consider citizen i with type $(\theta_i, \tau_i) = (t(\tau'), \tau')$ for some $\tau' \in [\underline{\tau}, \bar{\tau}]$.

Optimality of t requires that expected welfare conditional on i being of type $(t(\tau'), \tau')$ is the same no matter whether i chooses $p_i = 0$ or $p_i = 1$: If this was not the case, say for concreteness $p_i = 1$ lead to a higher expected consumer welfare, then t could not be optimal: As the setup is continuous, it would then also be better for expected consumer surplus if i chose $p_i = 1$ if he was any type in an $\varepsilon > 0$ neighborhood of $(t(\tau'), \tau')$. But as expected welfare is just the expectation of expected welfare conditional on i's type over i's type we get that an alternative strategy t' which is slightly lower than t around τ' leads to higher expected consumer welfare than t. This contradicts the definition of t. Consequently, expected welfare conditional on i being of type $(t(\tau'), \tau')$ has to be the same no matter whether i chooses $p_i = 0$ or $p_i = 1$.

We are now going to show that the just stated (necessary) optimality condition cannot be satisfied for any $t > 0$. However, it is trivially satisfied for t^p by the symmetry of the setup. We focus on citizen i with type $\theta_i = t(\tau_i) > 0$. If citizen i chooses $p_i = 1$ instead of $p_i = 0$, he will increase the probability that $p = 1$ by $1/n$. This can be interpreted as follows: choosing $p_i = 1$ instead of $p_i = 0$ leads with probability $1/n$ to a payoff of θ instead of a payoff of zero (for each citizen). Hence, choosing $p_i = 1$ is best for expected consumer welfare (conditional on i's type) if $\mathbb{E}[\theta|\theta_i] > 0$.[2] As we showed above, $\mathbb{E}[\theta|\theta_i] = \theta_i/(1+\sigma^2)$ which is strictly positive for all $\theta_i > 0$. It follows that $p_i = 1$ leads to strictly higher expected consumer welfare than $p_i = 0$ as $\theta_i > 0$. This contradicts that $t > 0$ maximizes expected consumer surplus. □

The previous results can now be used to obtain a stronger version of our welfare result in the paper. While the paper argued that expected aggregated consumer surplus is higher under privacy if n is large

[2] Note that conditioning on τ_i is immaterial as τ_i is – given θ_i – not correlated with θ.

(while OP's payoff is the same with and without privacy), we can now say that the expected utility of each citizen – regardless of his type (θ_i, τ_i) – is higher under privacy for n large. That is, privacy is an interim Pareto improvement here while it was only an ex ante Pareto improvement in the model of the paper.

Proposition 2. *Assume OP plays M in the privacy equilibrium.*
1.) If OP uses a mixed – that is not pure – strategy in the equilibrium without privacy, then changing to the privacy case increases expected welfare at the interim stage.
2.) Assume that (i) δ is differentiable and strictly increasing in τ, i.e. $\delta'(\tau) > 0$ for all $\tau \in [\underline{\tau}, \bar{\tau}]$ and (ii) $\Gamma_\infty = \lim_{\theta_i \to \infty} \Gamma_{\theta_i}$ is a non-degenerate distribution in the sense that $\Gamma_\infty(\tau_i) > 0$ for all $\tau_i > \underline{\tau}$. Then, privacy welfare dominates no privacy for large n in the following sense: Compared to the no privacy case, privacy leads to a higher expected consumer surplus for each consumer of every type and the same expected payoff for OP if n is sufficiently large.

In order to prove the proposition, we have to first restate the technical result on the limit of tails of Φ that we show in the appendix for $\hat{\Phi}(\theta_i)$.

Lemma 4. *Let $\hat{\Phi}(\theta_i) = \int_{\mathbb{R}} \tilde{\Phi}_\theta(\theta_i)\, d\Phi(\theta)$ be the distribution of θ_i from OP's perspective. Then, $\int_{ka-b}^{ka} d\hat{\Phi} / \int_{ka}^{\infty} d\hat{\Phi}$ diverges to infinity as $k \to \infty$ for $a, b > 0$.*

Proof. If we can show that $\hat{\phi}(x)/\hat{\phi}(x+b)$ diverges to infinity as $x \to \infty$ (where $\hat{\phi}$ is the density of $\hat{\Phi}$), then the same proof as in the paper applies. Note that

$$\hat{\phi}(x) = \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\theta)^2}{2\sigma^2}} d\Phi(\theta) = \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\theta)^2}{2\sigma^2}} e^{-\theta^2/2}\, d\theta$$

$$\begin{aligned}
\frac{\hat{\phi}(x)}{\hat{\phi}(x+b)} &= \frac{\int_{\mathbb{R}} e^{-(x-\theta)^2/(2\sigma^2)}\, d\Phi(\theta)}{\int_{\mathbb{R}} e^{-(x+b-\theta)^2/(2\sigma^2)}\, d\Phi(\theta)} \\
&= \frac{\int_{\mathbb{R}} e^{-(x-\theta)^2/(2\sigma^2)-\theta^2/2}\, d\theta}{\int_{\mathbb{R}} e^{-(x+b-\theta)^2/(2\sigma^2)-\theta^2/2}\, d\theta} \\
&= \frac{\int_{\mathbb{R}} e^{[-(1+\sigma^2)\theta^2+2x\theta]/(2\sigma^2)}\, d\theta}{\int_{\mathbb{R}} e^{[-2b(x-\theta)-b^2-(1+\sigma^2)\theta^2+2x\theta]/(2\sigma^2)}\, d\theta} \\
&= \frac{\int_{\mathbb{R}} e^{-\frac{(\theta-x/(1+\sigma^2))^2}{2\sigma^2/(1+\sigma^2)}}\, d\theta}{\int_{\mathbb{R}} e^{(-2b(x-\theta)-b^2)/(2\sigma^2)} e^{-\frac{(\theta-x/(1+\sigma^2))^2}{2\sigma^2/(1+\sigma^2)}}\, d\theta} \\
&= \frac{\int_{\mathbb{R}} d\bar{\Phi}(\theta)}{\int_{\mathbb{R}} e^{(-2b(x-\theta)-b^2)/(2\sigma^2)}\, d\bar{\Phi}(\theta)}
\end{aligned}$$

where $\bar{\Phi}$ is the cdf of a normal distribution with mean $x/(1+\sigma^2)$ and variance $\sigma^2/(1+\sigma^2)$. As the numerator is 1, the previous expression can be written as

$$\frac{\hat{\phi}(x)}{\hat{\phi}(x+b)} = \frac{1}{\int_{\mathbb{R}} e^{-\frac{2b(x-\theta)+b^2}{2\sigma^2}}\, d\bar{\Phi}(\theta)}$$

which diverges to infinity as $x \to \infty$ (because the denominator converges to zero). Given this, the rest of the proof from the main paper goes through one-to-one which implies the lemma. □

Proof of proposition 2: Let M be optimal for OP in the privacy equilibrium.

1.) Suppose there is a mixed strategy equilibrium in the case without privacy. Then, OP has to play M against both groups with positive probability. If he played A against those who chose $p_i = 1$ for sure and mixed for those who chose $p_i = 0$, then M could not be optimal in the privacy case. Hence, OP can

in the case without privacy achieve a payoff equal to his equilibrium payoff by playing M against both groups. Consequently, OP's payoff with and without privacy is the same. Citizens are strictly better off with privacy as (a) there is no chilling effect which means by lemma 3 that expected welfare of every consumer (no matter which type) in the information aggregation stage is maximized and (b) M will be played with probability 1 against them in the interaction stage.

2.) Now assume that $\delta'(\tau) > 0$. We will show that for n sufficiently high the privacy equilibrium welfare dominates the equilibrium in the case without privacy (or the two are identical).

We are going to make use of the fact that for any $\mu > 0$, we can find an $\hat{n}$ so that for all $n > \hat{n}$, $q((m+1)/n) - q(m/n) = 1/n < \mu$. Now recall that $t^{np}(\tau_i) = (1+\sigma^2)n\delta(\tau_i)\Delta$. Consequently, the threshold values become arbitrarily large as n gets large (assuming $\Delta = 1$). Note also that t is increasing in τ and the slope also becomes arbitrarily large as n increases. From here, the proof of the main paper applies with $\hat{\Phi}$ in place of Φ. $\square$